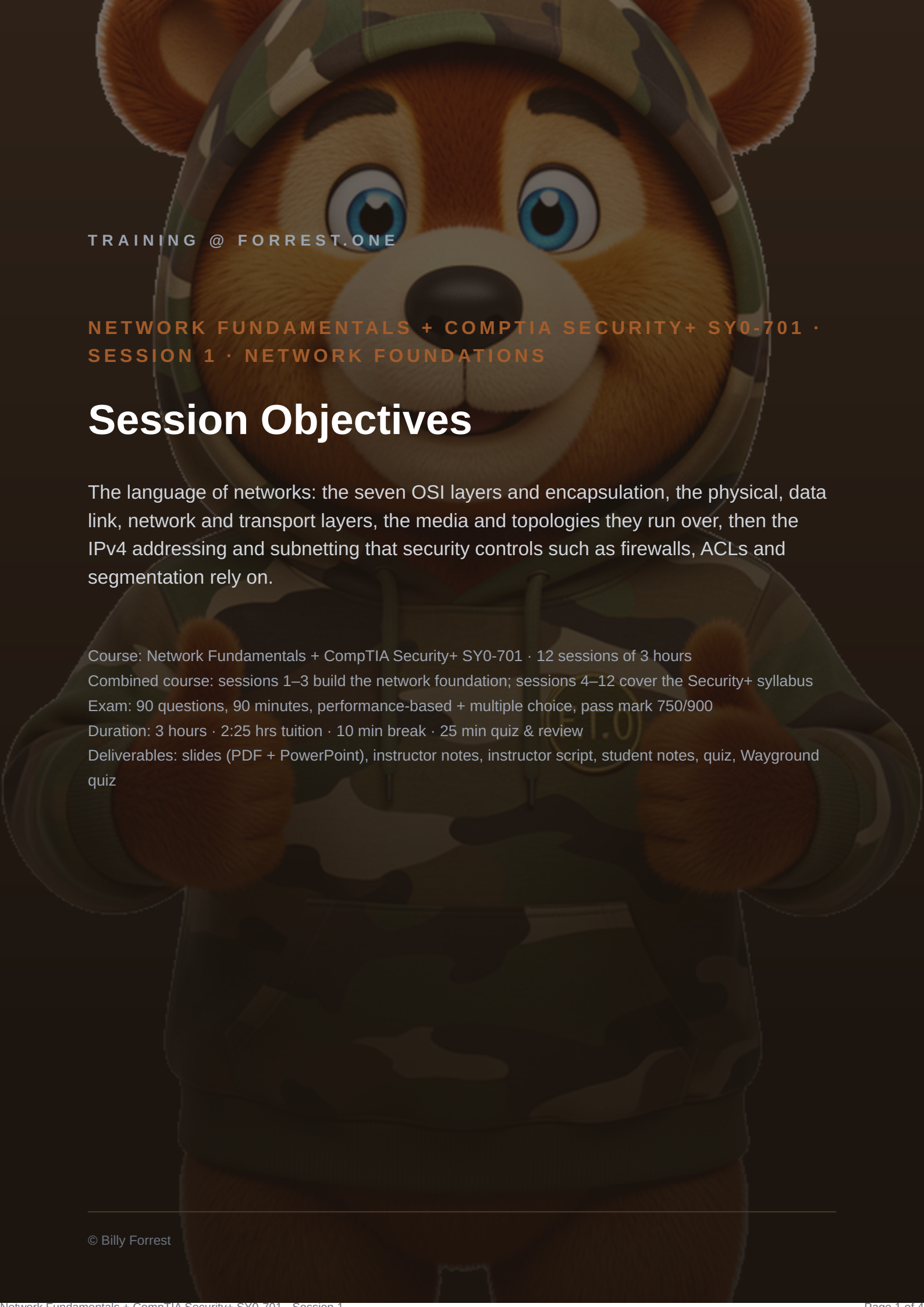




TRAINING @ FORREST.ONE

**NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 1 · NETWORK FOUNDATIONS**

Session Objectives

The language of networks: the seven OSI layers and encapsulation, the physical, data link, network and transport layers, the media and topologies they run over, then the IPv4 addressing and subnetting that security controls such as firewalls, ACLs and segmentation rely on.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Combined course: sessions 1–3 build the network foundation; sessions 4–12 cover the Security+ syllabus

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, quiz, Wayground quiz

Session Objectives — Network Foundations: Networking Models & IP Addressing

This first session of the combined Network Fundamentals + CompTIA Security+ SY0-701 course builds the network foundation that every later security session depends on. It opens with why we layer networks at all, then walks the seven OSI layers one at a time: the physical layer and its signals, cabling and connectors; the data link layer, its two sublayers and hardware addressing; the network and transport layers, with Time to Live and the choice between Transmission Control Protocol and User Datagram Protocol; and the session, presentation and application layers. It then covers the media signals travel over, wired and wireless, together with the topologies we arrange them in. Finally it covers IPv4 addressing, subnet masks, classless prefixes, the private and reserved ranges, and the prefix table. Throughout, the security angle stays live: layer awareness as the way to choose a control, hardware addresses as untrustworthy identity, and subnetting as the geometry of segmentation. It requires no prior networking knowledge.

Learning outcomes

By the end of the 3-hour session, candidates will be able to:

- Explain why networks are organised into layers, and why that makes faults findable.
- Name all seven OSI layers and state the job each one performs.
- Describe the journey of a message down the stack, naming what each layer adds and which address it uses.
- Explain encapsulation and name the unit of data at each layer: data, segment or datagram, packet, frame, bits.
- Describe the physical layer, including its three signal types and the cabling, connector and signalling standards it defines.
- State why a completely dead link is almost always a layer 1 fault and what to check first.
- Explain Media Access Control addressing: 48 bits, hexadecimal notation, and why a hardware address is not a trustworthy identity.
- Distinguish the two data link sublayers: Logical Link Control and Media Access Control, and state which layer each faces.
- Describe the network layer: logical addressing, routers, Time to Live, and route adaptation when a link fails.
- Compare Transmission Control Protocol and User Datagram Protocol on connection, reliability, ordering, speed and typical use.
- Describe the session, presentation and application layers, and correct the common error that the application layer is the program.

- Choose wired and wireless media for a given requirement, using distance, bandwidth and interference as the deciding factors.
- Recognise the common connectors and transceivers: Registered Jack 45, Subscriber and Lucent Connectors, Small Form factor Pluggable modules, media converters.
- Compare star, mesh and hybrid topologies, and state the cost, resilience and single point of failure trade-offs of each.
- Read an IPv4 address and subnet mask, and derive the network address, usable host range and broadcast address.
- Recognise the private, loopback and automatic address ranges on sight, and interpret a 169.254 address correctly.
- Convert between prefix notation and subnet masks, and recall the usable host counts from /24 down to /30.
- Explain how classless addressing replaced the class system, and why a larger prefix means a smaller network.
- Describe VLSM as per-subnet sizing allocated largest first, and explain how subnet design acts as a security control by limiting lateral movement.

Course context

Item	Detail
Course	Network Fundamentals + CompTIA Security+ SY0-701, 12 sessions of 3 hours
Strand	Network foundation (sessions 1–3)
Prerequisites	Basic binary numbering; no prior networking knowledge assumed
Assessment	HTML learning quiz (20 questions) + Wayground quiz (15 questions)
Exam link	Security+ Domain 3 architecture and Domain 4 operations assume this material

Session 1 coverage

Chapter	Title	What is covered
1	The Language of Networks	Why layering exists at all; troubleshooting one layer at a time
2	The OSI Model	The seven layers and their jobs; the journey of a message; encapsulation and the unit of data at each layer
3	Physical and Data Link	Signals, media and standards; layer 1 as the first troubleshooting check; Media Access Control addressing; Logical Link Control against Media Access Control; Cyclic Redundancy Check; address spoofing
4	Network, Transport and Above	

Chapter	Title	What is covered
		Logical addressing, routers and Time to Live; ports and the TCP/UDP comparison; session, presentation and application layers; naming the layer to choose the control
5	Media and Topologies	Wireless transmission and frequency trade-offs; twisted pair, shielded twisted pair, coaxial and fibre; connectors and transceivers; star, mesh, hybrid and spine and leaf
6	IPv4 Addressing and Subnetting	32-bit structure and dotted decimal; private, loopback and automatic ranges; subnet masks and reserved addresses; worked mask example; classless addressing and VLSM; the prefix-against-hosts table

How this session maps to the course

Sessions 1–3 give you the network vocabulary. Session 5 (threats), Session 8 (enterprise network security) and Session 11 (security operations) all assume you can name a layer, read an address and size a subnet. Every quiz question is tagged with its topic area, such as 1B IPv4 Addressing.